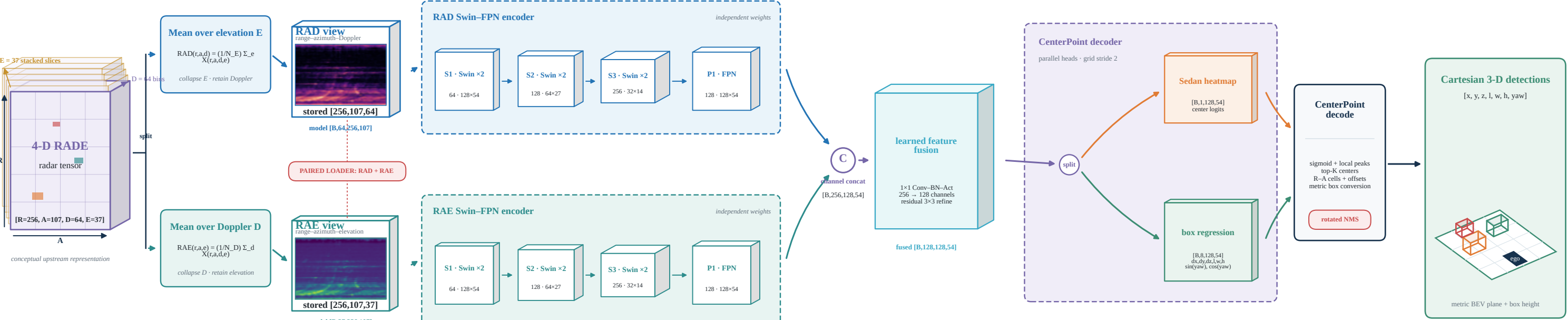


RADE-to-dual-view processing and Model7 detection overview

Upstream tensor projection → independent Swin-FPN encoders → encoded-feature fusion → dense Cartesian detection

- 14-D RADAR TENSOR
- 2MEAN PROJECTIONS + STORED VIEWS
- 3INDEPENDENT SWIN-FPN ENCODERS
- 4ENCODED CONCAT + FUSION
- 5DENSE DECODER HEADS
- 6DECODE
- 7OUTPUT



Shallow 3-D drawing convention

RADE: front = $R \times A$ · cube depth = D · repeated slices = E

Features: front = spatial grid · depth = channels

Perspective is illustrative; it does not denote 3-D convolution.



Implementation boundary and provenance

Upstream: RADE is projected into views. Active loader: paired precomputed RAD/RAE .npy tensors are read directly.

Model7 concatenates encoded Swin-FPN features—not raw views. The endpoint here is 3-D detection, not semantic segmentation.

Overview convention inspired by Ouaknine et al., Multi-View Radar Semantic Segmentation, ICCV 2021 Fig. 1 / thesis Fig. 5.1; graphics are original.